

Empirical Processes, Uniform Convergence, and Complexity

SDS8102 Mathematical Statistics

1. From the Law of Large Numbers to Empirical Processes

Let

$$X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} P$$

on a measurable space $(X, \mathcal{A})$.

For a measurable function $f : X \rightarrow \mathbb{R}$, define

$$Pf := \int f \, dP = \mathbb{E}[f(X)]$$

and

$$P_n f := \frac{1}{n} \sum_{i=1}^n f(X_i).$$

For a fixed integrable f , the law of large numbers gives

$$P_n f \longrightarrow Pf.$$

In many statistical problems, however, we need this approximation to hold simultaneously over a large collection of functions.

Given a class $\mathcal{F}$, the central object is therefore

$$\sup_{f \in \mathcal{F}} |P_n f - Pf|.$$

The fundamental question of empirical process theory is:

How complicated can $\mathcal{F}$ be while the empirical measure still approximates the population measure uniformly over $\mathcal{F}$?

This question appears throughout statistics and machine learning: empirical distribution functions, nonparametric estimation, classification, empirical risk minimization, confidence bands, M-estimation, and many other problems.

1.1. The empirical measure

Definition 1.1 (Empirical measure). The empirical measure associated with $X_1, \dots, X_n$ is

$$P_n = \frac{1}{n} \sum_{i=1}^n \delta_{X_i},$$

where δ_x denotes the point mass at x .

For any measurable f ,

$$P_n f = \int f \, dP_n = \frac{1}{n} \sum_{i=1}^n f(X_i).$$

Thus P_n is a random approximation to P .

1.2. The empirical process

Definition 1.2 (Empirical process). The centered and scaled empirical process is

$$\mathbb{G}_n f := \sqrt{n}(P_n - P)f = \frac{1}{\sqrt{n}} \sum_{i=1}^n \{f(X_i) - Pf\}, \quad f \in \mathcal{F}.$$

There are two natural scales.

The first is the law-of-large-numbers scale:

$$\sup_{f \in \mathcal{F}} |(P_n - P)f|.$$

The second is the central-limit scale:

$$\sup_{f \in \mathcal{F}} |\mathbb{G}_n f| = \sqrt{n} \sup_{f \in \mathcal{F}} |(P_n - P)f|.$$

This leads to two fundamental theories:

Uniform LLN	$\longleftrightarrow$	Glivenko–Cantelli theory,
Uniform CLT	$\longleftrightarrow$	Donsker theory.

2. The Empirical CDF: The Canonical Example

Suppose now that $X_i \in \mathbb{R}$ with distribution function

$$F(t) = P(X \leq t).$$

The empirical distribution function is

$$F_n(t) = \frac{1}{n} \sum_{i=1}^n \mathbf{1}\{X_i \leq t\}.$$

Define

$$f_t(x) = \mathbf{1}\{x \leq t\}.$$

Then

$$F_n(t) = P_n f_t, \quad F(t) = P f_t.$$

Hence

$$F_n(t) - F(t) = (P_n - P)f_t.$$

The entire empirical CDF is therefore an empirical process indexed by the class

$$\mathcal{F} = \{f_t : t \in \mathbb{R}\} = \{\mathbf{1}_{(-\infty, t]} : t \in \mathbb{R}\}.$$

For each fixed t , the strong law gives

$$F_n(t) \rightarrow F(t) \quad \text{a.s.}$$

But the classical Glivenko–Cantelli theorem says much more.

3. The Glivenko–Cantelli Theorem

Theorem 3.1 (Glivenko–Cantelli). *Let F_n be the empirical CDF based on an i.i.d. sample from distribution function F . Then*

$$\sup_{t \in \mathbb{R}} |F_n(t) - F(t)| \longrightarrow 0 \quad \text{a.s.}$$

The important word is *uniformly*. The strong law gives pointwise convergence at each fixed t . Glivenko–Cantelli upgrades this to simultaneous convergence over all $t \in \mathbb{R}$.

Proof. Fix $\varepsilon > 0$.

Choose points

$$-\infty = t_0 < t_1 < \cdots < t_m = \infty$$

such that

$$F(t_j) - F(t_{j-1}) \leq \varepsilon$$

for each j , with the partition chosen at appropriate quantile levels.

For any $t_{j-1} \leq t \leq t_j$, monotonicity gives

$$F_n(t_{j-1}) \leq F_n(t) \leq F_n(t_j)$$

and

$$F(t_{j-1}) \leq F(t) \leq F(t_j).$$

Consequently,

$$F_n(t) - F(t) \leq F_n(t_j) - F(t_{j-1}).$$

Write

$$F_n(t_j) - F(t_{j-1}) = \{F_n(t_j) - F(t_j)\} + \{F(t_j) - F(t_{j-1})\},$$

so

$$F_n(t) - F(t) \leq |F_n(t_j) - F(t_j)| + \varepsilon.$$

Similarly,

$$F(t) - F_n(t) \leq |F_n(t_{j-1}) - F(t_{j-1})| + \varepsilon.$$

Therefore

$$\sup_t |F_n(t) - F(t)| \leq \max_{0 \leq j \leq m} |F_n(t_j) - F(t_j)| + \varepsilon.$$

There are only finitely many t_j . By the strong law,

$$F_n(t_j) \rightarrow F(t_j) \quad \text{a.s.}$$

simultaneously for all j . Hence

$$\limsup_{n \rightarrow \infty} \sup_t |F_n(t) - F(t)| \leq \varepsilon \quad \text{a.s.}$$

Since $\varepsilon > 0$ is arbitrary,

$$\sup_t |F_n(t) - F(t)| \rightarrow 0 \quad \text{a.s.}$$

■

4. Glivenko–Cantelli Classes

The classical theorem suggests the following abstraction.

Definition 4.1 (Glivenko–Cantelli class). A class $\mathcal{F}$ of measurable functions is called P -Glivenko–Cantelli if

$$\sup_{f \in \mathcal{F}} |P_n f - P f| \longrightarrow 0 \quad \text{a.s.}$$

Thus the class of half-lines

$$\{\mathbf{1}_{(-\infty, t]} : t \in \mathbb{R}\}$$

is Glivenko–Cantelli for every probability distribution on $\mathbb{R}$.

Remark 4.2. For uncountable classes $\mathcal{F}$, the supremum need not automatically be measurable. A fully general treatment uses separability or outer probability and outer expectation. We suppress these technicalities and assume throughout that the relevant suprema are measurable.

4.1. Why pointwise convergence is insufficient

For every fixed $f \in \mathcal{F}$,

$$P_n f \rightarrow P f.$$

But this does not imply

$$\sup_{f \in \mathcal{F}} |P_n f - P f| \rightarrow 0.$$

The issue is that the function attaining the supremum may depend on the data and on n . Uniform convergence therefore requires some control on the *complexity* of $\mathcal{F}$.

4.2. Finite classes

The simplest example illustrates the basic principle.

Suppose

$$\mathcal{F} = \{f_1, \dots, f_N\}$$

and every $f \in \mathcal{F}$ satisfies $|f| \leq 1$.

Hoeffding's inequality gives, for each fixed f ,

$$\mathbb{P}(|P_n f - P f| > t) \leq 2e^{-nt^2/2}.$$

By the union bound,

$$\mathbb{P}\left(\sup_{f \in \mathcal{F}} |P_n f - P f| > t\right) \leq 2Ne^{-nt^2/2}.$$

Thus the typical uniform deviation is of order

$$\sqrt{\frac{\log N}{n}}.$$

The relevant measure of complexity is therefore not N itself, but roughly

$$\log N.$$

For infinite classes we need a suitable replacement for $\log |\mathcal{F}|$.

5. Symmetrization

Directly studying

$$\sup_{f \in \mathcal{F}} |P_n f - P f|$$

is difficult because $P f$ is an unknown population quantity.

The key idea is to replace the population expectation by an independent empirical average.

Let

$$X'_1, \dots, X'_n$$

be an independent copy of the original sample and define

$$P'_n f = \frac{1}{n} \sum_{i=1}^n f(X'_i).$$

5.1. Ghost-sample symmetrization

By Jensen's inequality,

$$\begin{aligned} \mathbb{E} \sup_{f \in \mathcal{F}} |P_n f - P f| &= \mathbb{E} \sup_{f \in \mathcal{F}} |P_n f - \mathbb{E}[P'_n f \mid X_1, \dots, X_n]| \\ &\leq \mathbb{E} \sup_{f \in \mathcal{F}} |P_n f - P'_n f|. \end{aligned}$$

Thus

$$\mathbb{E} \sup_{f \in \mathcal{F}} |P_n f - P f| \leq \mathbb{E} \sup_{f \in \mathcal{F}} \left| \frac{1}{n} \sum_{i=1}^n \{f(X_i) - f(X'_i)\} \right|.$$

The population distribution has disappeared.

5.2. Rademacher randomization

Let

$$\varepsilon_1, \dots, \varepsilon_n$$

be independent Rademacher variables:

$$\mathbb{P}(\varepsilon_i = 1) = \mathbb{P}(\varepsilon_i = -1) = \frac{1}{2}.$$

Because (X_i, X'_i) has the same distribution as (X'_i, X_i) ,

$$f(X_i) - f(X'_i)$$

has the same joint distribution after multiplying independently by ε_i .

Therefore

$$\begin{aligned} \mathbb{E} \sup_{f \in \mathcal{F}} |P_n f - P'_n f| &= \mathbb{E} \sup_{f \in \mathcal{F}} \left| \frac{1}{n} \sum_{i=1}^n \varepsilon_i \{f(X_i) - f(X'_i)\} \right| \\ &\leq \mathbb{E} \sup_{f \in \mathcal{F}} \left| \frac{1}{n} \sum_{i=1}^n \varepsilon_i f(X_i) \right| \\ &\quad + \mathbb{E} \sup_{f \in \mathcal{F}} \left| \frac{1}{n} \sum_{i=1}^n \varepsilon_i f(X'_i) \right|. \end{aligned}$$

The two terms have the same distribution.

Hence we obtain the fundamental symmetrization inequality.

Theorem 5.1 (Symmetrization). *Let $\mathcal{F}$ be a class of integrable measurable functions. Then*

$$\mathbb{E} \sup_{f \in \mathcal{F}} |P_n f - P f| \leq 2 \mathbb{E} \sup_{f \in \mathcal{F}} \left| \frac{1}{n} \sum_{i=1}^n \varepsilon_i f(X_i) \right|.$$

The complicated empirical process has been replaced by a process with random signs.

6. Rademacher Complexity

Definition 6.1 (Empirical Rademacher complexity). Given $X_1, \dots, X_n$, define

$$\widehat{\mathfrak{R}}_n(\mathcal{F}) = \mathbb{E}_{\varepsilon} \left[\sup_{f \in \mathcal{F}} \left| \frac{1}{n} \sum_{i=1}^n \varepsilon_i f(X_i) \right| \middle| X_1, \dots, X_n \right].$$

The expected Rademacher complexity is

$$\mathfrak{R}_n(\mathcal{F}) = \mathbb{E}_X \widehat{\mathfrak{R}}_n(\mathcal{F}).$$

Symmetrization gives immediately

$$\mathbb{E} \sup_{f \in \mathcal{F}} |P_n f - P f| \leq 2 \mathfrak{R}_n(\mathcal{F}).$$

Thus Rademacher complexity measures the effective size of $\mathcal{F}$.

A useful interpretation is:

How well can functions in $\mathcal{F}$ correlate with completely random signs?

A class that can fit random noise is rich. A class that cannot fit random noise has low effective complexity.

6.1. A finite-class bound

Suppose

$$\mathcal{F} = \{f_1, \dots, f_N\}$$

and

$$|f_j(X_i)| \leq B.$$

Conditional on the sample,

$$Z_f = \frac{1}{n} \sum_{i=1}^n \varepsilon_i f(X_i)$$

is sub-Gaussian.

Indeed,

$$\mathbb{E}_{\varepsilon} e^{\lambda Z_f} = \prod_{i=1}^n \mathbb{E}_{\varepsilon} \exp \left(\frac{\lambda \varepsilon_i f(X_i)}{n} \right) \leq \exp \left(\frac{\lambda^2}{2n^2} \sum_{i=1}^n f(X_i)^2 \right).$$

Hence standard maximal inequalities for sub-Gaussian random variables give

$$\widehat{\mathfrak{R}}_n(\mathcal{F}) \lesssim B \sqrt{\frac{\log(2N)}{n}}.$$

Again complexity appears through the logarithm of the number of distinguishable functions.

7. Application: Generalization and Empirical Risk Minimization

Let $Z = (X, Y)$ denote an observation and let $\mathcal{F}$ be a class of prediction rules.

For $f \in \mathcal{F}$, define the population risk

$$R(f) = \mathbb{E}[\ell(f(X), Y)]$$

and empirical risk

$$\widehat{R}_n(f) = \frac{1}{n} \sum_{i=1}^n \ell(f(X_i), Y_i).$$

Let

$$\mathcal{L} = \{z \mapsto \ell(f(x), y) : f \in \mathcal{F}\}$$

be the induced loss class.

Then

$$\sup_{f \in \mathcal{F}} |\widehat{R}_n(f) - R(f)| = \sup_{g \in \mathcal{L}} |P_n g - P g|.$$

Thus generalization is precisely an empirical-process problem.

7.1. Excess risk of empirical risk minimization

Let

$$\widehat{f} \in \arg \min_{f \in \mathcal{F}} \widehat{R}_n(f),$$

and let

$$f^* \in \arg \min_{f \in \mathcal{F}} R(f).$$

Then

$$\begin{aligned} R(\widehat{f}) - R(f^*) &= R(\widehat{f}) - \widehat{R}_n(\widehat{f}) \\ &\quad + \widehat{R}_n(\widehat{f}) - \widehat{R}_n(f^*) \\ &\quad + \widehat{R}_n(f^*) - R(f^*). \end{aligned}$$

Since $\widehat{f}$ minimizes empirical risk,

$$\widehat{R}_n(\widehat{f}) - \widehat{R}_n(f^*) \leq 0.$$

Therefore

$$R(\widehat{f}) - R(f^*) \leq 2 \sup_{f \in \mathcal{F}} |\widehat{R}_n(f) - R(f)|.$$

Hence

uniform convergence $\implies$ consistency of ERM.
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More quantitatively,

$$\mathbb{E}[R(\widehat{f}) - R(f^*)] \leq 4\mathfrak{R}_n(\mathcal{L}).$$

8. Example: Linear Prediction

Consider

$$\mathcal{F} = \{x \mapsto w^\top x : \|w\|_2 \leq B\},$$

and suppose

$$\|X_i\|_2 \leq R.$$

Then

$$\begin{aligned} \widehat{\mathfrak{R}}_n(\mathcal{F}) &= \mathbb{E}_\varepsilon \sup_{\|w\| \leq B} \left| \frac{1}{n} \sum_{i=1}^n \varepsilon_i w^\top X_i \right| \\ &= \frac{1}{n} \mathbb{E}_\varepsilon \sup_{\|w\| \leq B} \left| w^\top \sum_{i=1}^n \varepsilon_i X_i \right|. \end{aligned}$$

By Cauchy–Schwarz,

$$\sup_{\|w\| \leq B} |w^\top v| = B \|v\|_2.$$

Thus

$$\widehat{\mathfrak{R}}_n(\mathcal{F}) = \frac{B}{n} \mathbb{E}_\varepsilon \left\| \sum_{i=1}^n \varepsilon_i X_i \right\|_2.$$

By Jensen’s inequality,

$$\mathbb{E}_\varepsilon \left\| \sum_i \varepsilon_i X_i \right\|_2 \leq \left[\mathbb{E}_\varepsilon \left\| \sum_i \varepsilon_i X_i \right\|_2^2 \right]^{1/2}.$$

Since

$$\mathbb{E}[\varepsilon_i \varepsilon_j] = 0 \quad (i \neq j),$$

we obtain

$$\mathbb{E}_\varepsilon \left\| \sum_i \varepsilon_i X_i \right\|_2^2 = \sum_{i=1}^n \|X_i\|_2^2.$$

Therefore

$$\widehat{\mathfrak{R}}_n(\mathcal{F}) \leq \frac{B}{n} \left(\sum_{i=1}^n \|X_i\|_2^2 \right)^{1/2} \leq \frac{BR}{\sqrt{n}}.$$

Although $\mathcal{F}$ is an infinite class, its Rademacher complexity decreases at the parametric rate $n^{-1/2}$.

9. The Contraction Principle

Rademacher complexity behaves well under Lipschitz transformations.

Theorem 9.1 (Contraction principle). *Suppose $\phi : \mathbb{R} \rightarrow \mathbb{R}$ is L -Lipschitz and $\phi(0) = 0$. Then*

$$\mathbb{E}_\varepsilon \sup_{f \in \mathcal{F}} \left| \sum_{i=1}^n \varepsilon_i \phi(f(X_i)) \right| \leq L \mathbb{E}_\varepsilon \sup_{f \in \mathcal{F}} \left| \sum_{i=1}^n \varepsilon_i f(X_i) \right|,$$

up to the conventional absolute constant depending on the precise form of the contraction inequality.

Consequently, if the loss is Lipschitz in its prediction argument, the complexity of the loss class is controlled by the complexity of the prediction class.

For example, for an L -Lipschitz loss,

$$\mathfrak{R}_n(\ell \circ \mathcal{F}) \lesssim L\mathfrak{R}_n(\mathcal{F}).$$

Thus for bounded linear predictors,

$$\mathfrak{R}_n(\ell \circ \mathcal{F}) \lesssim \frac{LBR}{\sqrt{n}}.$$

This directly yields a generalization bound.

10. Boolean Classes and VC Dimension

We now specialize to classes

$$\mathcal{H} \subseteq \{0, 1\}^{\mathcal{X}}.$$

Equivalently, $\mathcal{H}$ can be regarded as a collection of subsets of $\mathcal{X}$ through

$$h_A(x) = \mathbf{1}\{x \in A\}.$$

For points $x_1, \dots, x_n$, define the trace

$$\mathcal{H}|_{x_{1:n}} = \{(h(x_1), \dots, h(x_n)) : h \in \mathcal{H}\}.$$

This records all binary labelings of the sample that can be generated by $\mathcal{H}$.

10.1. Shattering

Definition 10.1 (Shattering). A finite set

$$\{x_1, \dots, x_m\}$$

is shattered by $\mathcal{H}$ if

$$\mathcal{H}|_{x_{1:m}} = \{0, 1\}^m.$$

Equivalently, $\mathcal{H}$ can realize all 2^m possible labelings of the points.

Definition 10.2 (VC dimension). The VC dimension of $\mathcal{H}$, denoted

$$\text{VC}(\mathcal{H}),$$

is the largest integer d for which some set of d points is shattered by $\mathcal{H}$.

If arbitrarily large finite sets can be shattered, we set

$$\text{VC}(\mathcal{H}) = \infty.$$

10.2. Examples

Example 10.3 (Thresholds). Consider

$$\mathcal{H} = \{\mathbf{1}\{x \leq t\} : t \in \mathbb{R}\}.$$

One point can be labeled either 0 or 1 by moving the threshold. Thus

$$\text{VC}(\mathcal{H}) \geq 1.$$

For two ordered points $x_1 < x_2$, the labeling

$$(0, 1)$$

cannot be realized by a left half-line.

Hence

$$\boxed{\text{VC}(\mathcal{H}) = 1.}$$

Example 10.4 (Intervals). Consider

$$\mathcal{H} = \{\mathbf{1}\{x \in [a, b]\} : a \leq b\}.$$

Any two points can be shattered.

But for three ordered points

$$x_1 < x_2 < x_3,$$

the labeling

$$(1, 0, 1)$$

cannot be generated by a single interval.

Therefore

$$\boxed{\text{VC}(\mathcal{H}) = 2.}$$

Example 10.5 (Halfspaces). Consider affine halfspaces in $\mathbb{R}^d$:

$$\mathcal{H} = \{x \mapsto \mathbf{1}\{w^\top x + b \geq 0\} : w \in \mathbb{R}^d, b \in \mathbb{R}\}.$$

Then

$$\boxed{\text{VC}(\mathcal{H}) = d + 1.}$$

11. The Growth Function

Definition 11.1 (Growth function). The growth function of $\mathcal{H}$ is

$$\Pi_{\mathcal{H}}(n) = \sup_{x_1, \dots, x_n} |\mathcal{H}|_{x_{1:n}}|.$$

Clearly,

$$\Pi_{\mathcal{H}}(n) \leq 2^n.$$

If $\mathcal{H}$ shatters some n points, then

$$\Pi_{\mathcal{H}}(n) = 2^n.$$

Finite VC dimension implies that this exponential growth must eventually stop. The remarkable fact is that the growth becomes polynomial.

12. Sauer–Shelah Lemma

Theorem 12.1 (Sauer–Shelah). *If*

$$\text{VC}(\mathcal{H}) = d < \infty,$$

then for $n \geq d$,

$$\Pi_{\mathcal{H}}(n) \leq \sum_{j=0}^d \binom{n}{j}.$$

Moreover,

$$\sum_{j=0}^d \binom{n}{j} \leq \left(\frac{en}{d}\right)^d.$$

Thus

$$\log \Pi_{\mathcal{H}}(n) \lesssim d \log \left(\frac{en}{d}\right).$$

This is the crucial combinatorial consequence of finite VC dimension.

Without a VC restriction, the number of labelings may be 2^n . With VC dimension d , the number is only polynomial in n .

12.1. Consequence for Rademacher complexity

Conditional on $X_1, \dots, X_n$, a Boolean class can produce at most

$$\Pi_{\mathcal{H}}(n)$$

distinct vectors

$$(h(X_1), \dots, h(X_n)).$$

Applying the finite-class Rademacher bound gives

$$\widehat{\mathfrak{R}}_n(\mathcal{H}) \lesssim \sqrt{\frac{\log \Pi_{\mathcal{H}}(n)}{n}}.$$

Using Sauer–Shelah,

$$\mathfrak{R}_n(\mathcal{H}) \lesssim \sqrt{\frac{d \log(en/d)}{n}}.$$

Symmetrization then yields

$$\mathbb{E} \sup_{h \in \mathcal{H}} |P_n h - Ph| \lesssim \sqrt{\frac{d \log(en/d)}{n}}.$$

In particular, if d is fixed,

$$\sup_{h \in \mathcal{H}} |P_n h - Ph| \rightarrow 0$$

in probability, and stronger arguments give almost-sure uniform convergence.

Thus finite VC dimension is a fundamental sufficient condition for a uniform law of large numbers.

12.2. Glivenko–Cantelli theorem for a bounded VC class

Let $X_1, X_2, \dots$ be i.i.d. with distribution P , and let $\mathcal{F}$ be a measurable class of functions satisfying

$$\sup_{f \in \mathcal{F}} \|f\|_\infty \leq 1.$$

Assume that $\mathcal{F}$ is a VC-subgraph class with finite VC dimension $V < \infty$. Define

$$P_n f = \frac{1}{n} \sum_{i=1}^n f(X_i).$$

Theorem 12.2 (Glivenko–Cantelli theorem). *Under the preceding assumptions,*

$$\sup_{f \in \mathcal{F}} |P_n f - P f| \longrightarrow 0 \quad \text{almost surely.}$$

Proof. Set

$$Z_n = \sup_{f \in \mathcal{F}} |P_n f - P f|.$$

By symmetrization,

$$\mathbb{E}[Z_n] \leq 2\mathbb{E} \left[\sup_{f \in \mathcal{F}} \left| \frac{1}{n} \sum_{i=1}^n \varepsilon_i f(X_i) \right| \right],$$

where $\varepsilon_1, \dots, \varepsilon_n$ are independent Rademacher random variables.

Since $\mathcal{F}$ is a bounded VC-subgraph class of dimension V , its covering numbers satisfy the VC entropy bound

$$\log N(\eta, \mathcal{F}, L_2(P_n)) \leq CV \log \left(\frac{C}{\eta} \right), \quad 0 < \eta \leq 1.$$

The corresponding Rademacher-complexity bound is

$$\mathbb{E} \left[\sup_{f \in \mathcal{F}} \left| \frac{1}{n} \sum_{i=1}^n \varepsilon_i f(X_i) \right| \right] \leq C \sqrt{\frac{V}{n}}.$$

Consequently,

$$\mathbb{E}[Z_n] \leq C \sqrt{\frac{V}{n}} \longrightarrow 0.$$

We now upgrade this expectation bound to almost-sure convergence. If one observation X_i is replaced by X'_i , then

$$|Z_n(X_1, \dots, X_i, \dots, X_n) - Z_n(X_1, \dots, X'_i, \dots, X_n)| \leq \sup_{f \in \mathcal{F}} \frac{|f(X_i) - f(X'_i)|}{n} \leq \frac{2}{n}.$$

Therefore, McDiarmid's inequality gives

$$P(Z_n - \mathbb{E}[Z_n] \geq t) \leq \exp \left\{ -\frac{nt^2}{2} \right\}.$$

Fix $\varepsilon > 0$. Since $\mathbb{E}[Z_n] \rightarrow 0$, for all sufficiently large n ,

$$\mathbb{E}[Z_n] \leq \frac{\varepsilon}{2}.$$

Hence

$$\begin{aligned} P(Z_n > \varepsilon) &\leq P\left(Z_n - \mathbb{E}[Z_n] > \frac{\varepsilon}{2}\right) \\ &\leq \exp\left\{-\frac{n\varepsilon^2}{8}\right\}. \end{aligned}$$

Since

$$\sum_{n=1}^{\infty} \exp\left\{-\frac{n\varepsilon^2}{8}\right\} < \infty,$$

the Borel–Cantelli lemma implies

$$\mathbb{P}(Z_n > \varepsilon \text{ infinitely often}) = 0.$$

Applying this to $\varepsilon = 1/k$, $k \geq 1$, yields

$$Z_n \longrightarrow 0 \quad \text{almost surely.}$$

Therefore,

$$\sup_{f \in \mathcal{F}} |P_n f - P f| \longrightarrow 0 \quad \text{almost surely.}$$

■

13. VC Dimension and Generalization

Consider binary classification with

$$Y \in \{0, 1\}$$

and classifiers

$$h : \mathcal{X} \rightarrow \{0, 1\}.$$

The classification risk is

$$R(h) = \mathbb{P}(h(X) \neq Y).$$

The empirical risk is

$$\widehat{R}_n(h) = \frac{1}{n} \sum_{i=1}^n \mathbf{1}\{h(X_i) \neq Y_i\}.$$

If the classifier class has VC dimension d , then its induced misclassification loss class also has controlled combinatorial complexity.

Consequently,

$$\sup_{h \in \mathcal{H}} |\widehat{R}_n(h) - R(h)|$$

is typically of order

$$\sqrt{\frac{d}{n}}$$

up to logarithmic factors under elementary VC bounds.

Hence ERM satisfies approximately

$$R(\widehat{h}) - \inf_{h \in \mathcal{H}} R(h) \lesssim \sqrt{\frac{d}{n}}.$$

The statistical interpretation is important:

larger VC dimension	$\implies$	larger estimation/generalization error.
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This is one mathematical formulation of the tradeoff between expressiveness and generalization.

14. Covering Numbers and Metric Entropy

VC dimension is especially natural for indicator classes. For more general function classes we need a geometric notion of complexity.

Definition 14.1 (Covering number). Let $(\mathcal{F}, d)$ be a pseudometric space.

An ε -net is a finite set

$$\{f_1, \dots, f_N\} \subseteq \mathcal{F}$$

such that for every $f \in \mathcal{F}$ there exists f_j satisfying

$$d(f, f_j) \leq \varepsilon.$$

The covering number

$$N(\varepsilon, \mathcal{F}, d)$$

is the minimum possible size of an ε -net.

The quantity

$$\log N(\varepsilon, \mathcal{F}, d)$$

is called the metric entropy.

It measures the effective number of distinguishable elements of $\mathcal{F}$ at resolution ε .

A finite class corresponds roughly to

$$\log N(\varepsilon, \mathcal{F}, d) \leq \log |\mathcal{F}|.$$

For an infinite class, the entropy depends on the resolution.

14.1. The empirical L_2 metric

For empirical processes, an especially natural metric is

$$\|f - g\|_{P_n, 2} = \left[\frac{1}{n} \sum_{i=1}^n \{f(X_i) - g(X_i)\}^2 \right]^{1/2}.$$

This metric is random.

That is not a defect: it adapts the complexity measure to the observed sample.

15. Why a Single Net Is Not Enough

Suppose we choose an ε -net $\mathcal{F}_\varepsilon$.

For every $f \in \mathcal{F}$, let $\pi_\varepsilon f$ denote a nearby element of the net.

Then

$$Z_f = Z_{\pi_\varepsilon f} + (Z_f - Z_{\pi_\varepsilon f}).$$

The first term involves only finitely many functions and can be controlled by a union bound.

But the second term is an approximation error.

A coarse net has few elements but a large approximation error.

A fine net has a small approximation error but may contain far too many elements.

Chaining solves this problem by using *many resolutions simultaneously*.

16. The Chaining Idea

Choose scales

$$\varepsilon_0 > \varepsilon_1 > \varepsilon_2 > \cdots \downarrow 0.$$

At scale ε_j , let $\mathcal{F}_j$ be an ε_j -net.

For each $f \in \mathcal{F}$, choose

$$\pi_j f \in \mathcal{F}_j$$

such that

$$d(f, \pi_j f) \leq \varepsilon_j.$$

Then

$$f = \pi_0 f + \sum_{j=1}^{\infty} (\pi_j f - \pi_{j-1} f),$$

at least in the metric-completion sense.

For a process Z_f ,

$$Z_f - Z_{\pi_0 f} = \sum_{j=1}^{\infty} (Z_{\pi_j f} - Z_{\pi_{j-1} f}).$$

Thus a large global fluctuation has been decomposed into fluctuations across progressively smaller scales. This is the central idea of chaining.

17. Sub-Gaussian Processes

Definition 17.1 (Sub-Gaussian increments). A stochastic process

$$\{Z_f : f \in \mathcal{F}\}$$

has sub-Gaussian increments with respect to d if

$$\mathbb{E} \exp\{\lambda(Z_f - Z_g)\} \leq \exp\left(\frac{\lambda^2 d(f, g)^2}{2}\right)$$

for all $f, g \in \mathcal{F}$ and $\lambda \in \mathbb{R}$.

Equivalently, one obtains a tail bound of the form

$$\mathbb{P}(|Z_f - Z_g| > t) \leq 2 \exp\left(-\frac{ct^2}{d(f, g)^2}\right).$$

The metric d therefore describes the natural geometry of the process.

18. Dudley's Chaining Bound

The chaining construction leads to one of the central maximal inequalities in empirical process theory.

Theorem 18.1 (Dudley's entropy bound). Suppose $\{Z_f : f \in \mathcal{F}\}$ is a centered process with sub-Gaussian increments with respect to d . Then, under suitable separability conditions,

$$\mathbb{E} \sup_{f \in \mathcal{F}} |Z_f - Z_{f_0}| \lesssim \int_0^{\text{diam}(\mathcal{F})} \sqrt{\log N(\varepsilon, \mathcal{F}, d)} \, d\varepsilon.$$

The quantity

$$\int_0^{\text{diam}(\mathcal{F})} \sqrt{\log N(\varepsilon, \mathcal{F}, d)} \, d\varepsilon$$

is called an entropy integral.

18.1. Heuristic derivation

Take

$$\varepsilon_j = 2^{-j} D, \quad D = \text{diam}(\mathcal{F}).$$

At scale j ,

$$|\mathcal{F}_j| \leq N(\varepsilon_j, \mathcal{F}, d).$$

Each increment

$$Z_{\pi_j f} - Z_{\pi_{j-1} f}$$

has sub-Gaussian scale of order ε_{j-1} .

The number of possible increments is controlled by roughly

$$|\mathcal{F}_j| |\mathcal{F}_{j-1}|.$$

The maximal inequality for finitely many sub-Gaussian variables gives a contribution at scale j of order

$$\varepsilon_j \sqrt{\log N(\varepsilon_j, \mathcal{F}, d)}.$$

Summing over scales gives

$$\mathbb{E} \sup_f |Z_f - Z_{f_0}| \lesssim \sum_{j \geq 0} \varepsilon_j \sqrt{\log N(\varepsilon_j, \mathcal{F}, d)}.$$

The dyadic sum is comparable to

$$\int_0^D \sqrt{\log N(\varepsilon, \mathcal{F}, d)} \, d\varepsilon.$$

The important point is that complexity is accumulated across *all scales*.

19. Chaining the Rademacher Process

Conditional on the sample, define

$$R_f = \frac{1}{n} \sum_{i=1}^n \varepsilon_i f(X_i).$$

For $f, g \in \mathcal{F}$,

$$R_f - R_g = \frac{1}{n} \sum_{i=1}^n \varepsilon_i \{f(X_i) - g(X_i)\}.$$

Using the Rademacher moment-generating-function bound,

$$\mathbb{E}_{\varepsilon} e^{\lambda(R_f - R_g)} \leq \exp \left[\frac{\lambda^2}{2n^2} \sum_{i=1}^n \{f(X_i) - g(X_i)\}^2 \right].$$

But

$$\frac{1}{n^2} \sum_{i=1}^n \{f(X_i) - g(X_i)\}^2 = \frac{1}{n} \|f - g\|_{P_n, 2}^2.$$

Thus the conditional Rademacher process is sub-Gaussian under

$$d_n(f, g) = \frac{1}{\sqrt{n}} \|f - g\|_{P_n, 2}.$$

Applying Dudley's bound yields

$$\widehat{\mathfrak{R}}_n(\mathcal{F}) \lesssim \frac{1}{\sqrt{n}} \int_0^{\text{diam}(\mathcal{F}, L_2(P_n))} \sqrt{\log N(\varepsilon, \mathcal{F}, L_2(P_n))} \, d\varepsilon.$$

This is one of the fundamental bridges in empirical process theory:

$$\text{uniform empirical error} \rightarrow \text{Rademacher process} \rightarrow \text{covering numbers} \rightarrow \text{chaining}.$$

20. VC Classes and Entropy

The combinatorial VC notion can be converted into a metric entropy bound.

For a class of sets $\mathcal{C}$, write

$$\mathcal{F} = \{\mathbf{1}_C : C \in \mathcal{C}\}.$$

For a probability measure Q ,

$$\|\mathbf{1}_A - \mathbf{1}_B\|_{L_2(Q)}^2 = Q(A \Delta B),$$

where $A \Delta B$ denotes the symmetric difference.

Finite VC dimension implies polynomial covering numbers.

Theorem 20.1 (VC entropy bound). *Let $\mathcal{F}$ be a uniformly bounded VC class with VC dimension d . Then there exist universal constants $A, C > 0$ such that*

$$\sup_Q N(\varepsilon \|F\|_{Q, 2}, \mathcal{F}, L_2(Q)) \leq \left(\frac{A}{\varepsilon}\right)^{Cd}, \quad 0 < \varepsilon < 1,$$

where F is an envelope for the class and the supremum is over probability measures Q .

Consequently,

$$\log N(\varepsilon, \mathcal{F}, L_2(Q)) \lesssim d \log(A/\varepsilon).$$

Plugging this into Dudley's integral gives

$$\int_0^1 \sqrt{d \log(A/\varepsilon)} \, d\varepsilon \lesssim \sqrt{d}.$$

Therefore

$$\mathfrak{R}_n(\mathcal{F}) \lesssim \sqrt{\frac{d}{n}}.$$

This improves the elementary Sauer–Shelah bound

$$\sqrt{\frac{d \log n}{n}}$$

by removing the unnecessary logarithmic factor.

This is a major payoff of chaining.

21. Glivenko–Cantelli via Complexity

The previous arguments give a general strategy.

Suppose

$$\mathfrak{R}_n(\mathcal{F}) \rightarrow 0.$$

Then symmetrization gives

$$\mathbb{E} \sup_{f \in \mathcal{F}} |P_n f - P f| \leq 2\mathfrak{R}_n(\mathcal{F}) \rightarrow 0.$$

Hence

$$\sup_{f \in \mathcal{F}} |P_n f - P f| \rightarrow 0$$

in probability.

With appropriate concentration and summability arguments this can often be strengthened to almost-sure convergence.

Thus

$$\boxed{\mathfrak{R}_n(\mathcal{F}) \rightarrow 0 \implies \text{uniform law of large numbers.}}$$

For VC classes with fixed dimension,

$$\mathfrak{R}_n(\mathcal{F}) \lesssim \sqrt{\frac{d}{n}} \rightarrow 0.$$

Thus bounded VC classes are fundamental examples of Glivenko–Cantelli classes.

22. From Uniform Laws to Uniform Central Limit Theorems

The Glivenko–Cantelli problem studies

$$(P_n - P)f.$$

We now study fluctuations at the CLT scale:

$$\mathbb{G}_n f = \sqrt{n}(P_n - P)f.$$

For each fixed f satisfying

$$P f^2 < \infty,$$

the ordinary central limit theorem gives

$$\mathbb{G}_n f \Rightarrow N(0, \sigma_f^2),$$

where

$$\sigma_f^2 = P(f - P f)^2.$$

For finitely many functions

$$f_1, \dots, f_k,$$

the multivariate CLT gives

$$(\mathbb{G}_n f_1, \dots, \mathbb{G}_n f_k) \Rightarrow N_k(0, \Sigma),$$

where

$$\Sigma_{ij} = P(f_i f_j) - P f_i P f_j.$$

This is called finite-dimensional convergence.

For an infinite class $\mathcal{F}$, finite-dimensional convergence alone is not enough.

We need to control the process simultaneously over all $f \in \mathcal{F}$.

23. Donsker Classes

Definition 23.1 (Donsker class). A class $\mathcal{F}$ is called P -Donsker if

$$\mathbb{G}_n \Rightarrow \mathbb{G}_P \quad \text{in } \ell^\infty(\mathcal{F}),$$

where $\mathbb{G}_P$ is a tight centered Gaussian process with covariance

$$\text{Cov}(\mathbb{G}_P f, \mathbb{G}_P g) = P(fg) - P f P g.$$

Thus:

$$\begin{array}{ll} \mathcal{F} \text{ is GC} & : \quad \sup_{f \in \mathcal{F}} |P_n f - P f| \rightarrow 0, \\ \mathcal{F} \text{ is Donsker} & : \quad \sqrt{n}(P_n - P) \text{ converges as a process.} \end{array}$$

The first is a uniform law of large numbers.

The second is a uniform central limit theorem.

24. The Classical Donsker Theorem

Return to

$$\mathcal{F} = \{\mathbf{1}_{(-\infty, t]} : t \in \mathbb{R}\}.$$

Then

$$\mathbb{G}_n f_t = \sqrt{n}\{F_n(t) - F(t)\}.$$

Theorem 24.1 (Donsker's theorem). *If F is continuous, then*

$$\sqrt{n}(F_n - F) \Rightarrow \mathbb{B} \circ F$$

in the space of bounded functions, where $\mathbb{B}$ is a standard Brownian bridge.

A Brownian bridge is a centered Gaussian process

$$\{\mathbb{B}(u) : 0 \leq u \leq 1\}$$

with covariance

$$\text{Cov}(\mathbb{B}(s), \mathbb{B}(t)) = \min(s, t) - st.$$

Indeed, for $s \leq t$,

$$\begin{aligned} \text{Cov}(\mathbf{1}\{X \leq s\}, \mathbf{1}\{X \leq t\}) &= P(X \leq s) - F(s)F(t) \\ &= F(s) - F(s)F(t). \end{aligned}$$

Thus the covariance of the empirical process agrees with

$$\min(F(s), F(t)) - F(s)F(t),$$

which is precisely the covariance of

$$\mathbb{B}(F(t)).$$

25. Why Donsker Is Harder Than the Ordinary CLT

For every fixed collection

$$f_1, \dots, f_k,$$

the multivariate CLT already gives the correct Gaussian limit.

The difficult part is preventing the process from oscillating too wildly as f varies.

Thus a Donsker theorem consists conceptually of two ingredients:

1. finite-dimensional convergence,
2. asymptotic tightness/equicontinuity.

The second part is where entropy and chaining enter.

26. The Natural Metric of the Gaussian Limit

For the limiting Gaussian process define

$$\rho_P(f, g)^2 = \mathbb{E}[(\mathbb{G}_P f - \mathbb{G}_P g)^2].$$

Using the covariance formula,

$$\rho_P(f, g)^2 = P \left[\{(f - g) - P(f - g)\}^2 \right].$$

In particular,

$$\rho_P(f, g) \leq \|f - g\|_{L_2(P)}.$$

Thus $L_2(P)$ covering numbers naturally control the geometry of the limiting Gaussian process.

This explains why the same metric entropy that appeared in Rademacher complexity also appears in Donsker theory.

27. An Entropy Criterion for Donsker Classes

A representative sufficient condition is the finiteness of an entropy integral.

Roughly, if $\mathcal{F}$ has a square-integrable envelope F and

$$\int_0^1 \sqrt{\log \sup_Q N(\varepsilon \|F\|_{Q,2}, \mathcal{F}, L_2(Q))} \, d\varepsilon < \infty,$$

together with appropriate measurability conditions, then $\mathcal{F}$ is P -Donsker.

The exact theorem requires some technical conditions, but the structural message is simple:

$$\text{finite entropy integral} \implies \text{uniform control of empirical-process fluctuations.}$$

Since VC classes satisfy polynomial entropy bounds,

$$N(\varepsilon, \mathcal{F}, L_2(Q)) \lesssim (A/\varepsilon)^{Cd},$$

their entropy integral is finite:

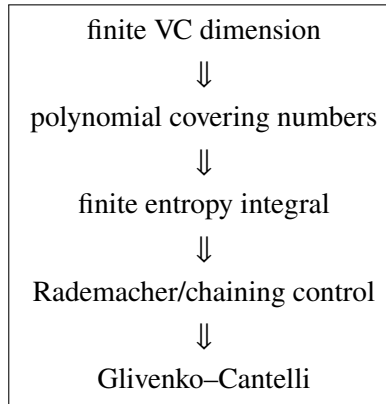
$$\int_0^1 \sqrt{d \log(A/\varepsilon)} \, d\varepsilon < \infty.$$

Hence suitably measurable, uniformly bounded VC classes are Donsker.

28. A Useful Hierarchy

We can now summarize the main complexity hierarchy.

For a sufficiently regular class $\mathcal{F}$,



and, under stronger envelope and measurability conditions,

entropy control	$\implies$	Donsker.
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The same notion of complexity therefore controls both uniform laws of large numbers and uniform central limit theorems.

29. Example: Thresholds Revisited

Consider

$$\mathcal{F} = \{\mathbf{1}\{x \leq t\} : t \in \mathbb{R}\}.$$

We have already seen that

$$\text{VC}(\mathcal{F}) = 1.$$

Therefore $\mathcal{F}$ has very small combinatorial complexity.

At the LLN scale,

$$\sup_t |F_n(t) - F(t)| \rightarrow 0 \quad \text{a.s.}$$

At the CLT scale,

$$\sqrt{n}(F_n - F) \Rightarrow \mathbb{B} \circ F.$$

Thus the classical Glivenko–Cantelli and Donsker theorems are the uniform LLN and uniform CLT for the same VC class.

30. Application: Kolmogorov–Smirnov Statistic

Suppose F is continuous.

The Kolmogorov–Smirnov statistic is

$$D_n = \sup_t |F_n(t) - F(t)|.$$

Glivenko–Cantelli tells us

$$D_n \rightarrow 0 \quad \text{a.s.}$$

But this gives no nondegenerate limiting distribution.
Donsker's theorem gives

$$\sqrt{n}D_n = \sup_t |\sqrt{n}(F_n(t) - F(t))| \Rightarrow \sup_{0 \leq u \leq 1} |\mathbb{B}(u)|.$$

Thus:

GC gives consistency; Donsker gives inference.

This distinction is fundamental throughout asymptotic statistics.

31. Application: Quantiles

Let

$$q_\alpha = F^{-1}(\alpha)$$

and define the empirical quantile

$$\widehat{q}_\alpha = F_n^{-1}(\alpha).$$

Uniform convergence of F_n to F gives consistency of empirical quantiles under appropriate regularity:

$$\widehat{q}_\alpha \rightarrow q_\alpha.$$

To obtain an asymptotic distribution, one needs the fluctuations of the empirical process.

If F has a positive continuous density f at q_α , then

$$\sqrt{n}(\widehat{q}_\alpha - q_\alpha) = -\frac{\sqrt{n}\{F_n(q_\alpha) - F(q_\alpha)\}}{f(q_\alpha)} + o_P(1).$$

Since

$$\sqrt{n}\{F_n(q_\alpha) - \alpha\} \Rightarrow N(0, \alpha(1 - \alpha)),$$

we obtain

$$\sqrt{n}(\widehat{q}_\alpha - q_\alpha) \Rightarrow N\left(0, \frac{\alpha(1 - \alpha)}{f(q_\alpha)^2}\right).$$

This is an early example of a general principle:

empirical process + linearization $\Rightarrow$ asymptotic distribution of an estimator.

32. GC versus Donsker

It is useful to keep the two theories conceptually separate.

A Glivenko–Cantelli statement says

$$\|P_n - P\|_{\mathcal{F}} = \sup_{f \in \mathcal{F}} |P_n f - P f| \rightarrow 0.$$

This establishes uniform consistency.

A Donsker statement says

$$\mathbb{G}_n = \sqrt{n}(P_n - P) \Rightarrow \mathbb{G}_P$$

as a random function indexed by $\mathcal{F}$.

This identifies the stochastic fluctuations around the population quantity.

Thus

$\begin{array}{ll} \text{GC} & \approx \text{uniform consistency,} \\ \text{Donsker} & \approx \text{uniform asymptotic distribution.} \end{array}$

A Donsker class is generally much more restrictive than merely being Glivenko–Cantelli.

33. The Main Proof Architecture

The main ideas developed in this chapter can be summarized as follows.

Start with the quantity

$$\mathbb{E} \sup_{f \in \mathcal{F}} |(P_n - P)f|.$$

Step 1: Symmetrization. Replace the unknown population mean by random signs:

$$\mathbb{E} \|P_n - P\|_{\mathcal{F}} \lesssim \mathbb{E} \sup_{f \in \mathcal{F}} \left| \frac{1}{n} \sum_i \varepsilon_i f(X_i) \right|.$$

Step 2: Identify the process metric. Conditional on the sample, the Rademacher process has sub-Gaussian increments under

$$d_n(f, g) = \frac{1}{\sqrt{n}} \|f - g\|_{P_n, 2}.$$

Step 3: Cover the function class. Measure the size of $\mathcal{F}$ through

$$N(\varepsilon, \mathcal{F}, L_2(P_n)).$$

Step 4: Chain across resolutions. Dudley’s inequality gives

$$\widehat{\mathfrak{R}}_n(\mathcal{F}) \lesssim \frac{1}{\sqrt{n}} \int \sqrt{\log N(\varepsilon, \mathcal{F}, L_2(P_n))} \, d\varepsilon.$$

Step 5: Bound the entropy. For a VC class,

$$\log N(\varepsilon, \mathcal{F}, L_2(Q)) \lesssim d \log(A/\varepsilon).$$

Step 6: Integrate. Therefore

$$\mathfrak{R}_n(\mathcal{F}) \lesssim \sqrt{\frac{d}{n}}.$$

This yields uniform convergence and, with stronger tightness arguments, functional central limit theorems.

34. Statistical Perspective

Empirical process theory extends the familiar LLN and CLT from one random variable to an entire collection of statistics.

For one fixed f ,

$$P_n f - P f$$

is an ordinary sample average.

For an entire class,

$$\{P_n f - P f : f \in \mathcal{F}\}$$

is a stochastic process.

The central difficulty is therefore not the behavior of any one function, but controlling all functions simultaneously.

The size of the class is measured in several complementary ways:

VC dimension	: combinatorial complexity,
Rademacher complexity	: ability to fit random noise,
covering numbers	: metric complexity at a given scale,
entropy integral	: complexity accumulated across scales.

These notions are connected by the chain

VC dimension $\longrightarrow$ covering numbers $\longrightarrow$ chaining $\longrightarrow$ Rademacher bounds $\longrightarrow$ uniform convergence.

At the law-of-large-numbers scale, this leads to Glivenko–Cantelli theory.

At the central-limit scale, it leads to Donsker theory.

In machine learning, exactly the same machinery controls generalization error.

Thus empirical process theory provides a common mathematical language for

probability + asymptotic statistics + statistical learning theory.
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35. Further Reading

For additional discussion of symmetrization, Rademacher complexity, VC dimension, and chaining, see the lecture notes listed on the course website. Particularly accessible introductions are the notes by Sen [1] and the treatment in van der Vaart [2]. A comprehensive reference for empirical process theory is van der Vaart and Wellner [3].

References

- [1] B. Sen. *A Gentle Introduction to Empirical Process Theory and Applications*. Lecture notes, 2018. <https://sites.stat.columbia.edu/bodhi/Talks/Emp-Proc-Lecture-Notes.pdf>.
- [2] A. W. van der Vaart. *Asymptotic Statistics*. Cambridge University Press, Cambridge, 1998.
- [3] A. W. van der Vaart and J. A. Wellner. *Weak Convergence and Empirical Processes: With Applications to Statistics*. Springer, New York, 1996.